

# NexaERP

*Business Process Analysis & ERP Solution Development*

[ GROUP No: 9 ]

## 1. Introduction

NexaERP is a comprehensive Enterprise Resource Planning solution designed for medium and largescale manufacturing and operational organizations. It addresses disconnected systems, manual processes, and delayed decision making by integrating Finance, Sales, Purchasing, Inventory, Manufacturing, HR, CRM, Project Management, Asset Management, and System Administration under one centralized database improving efficiency, minimizing errors, and enabling real-time, data driven decisionmaking.

## 2. Business Process Analysis

### 2.1 Industry Domain

NexaERP targets the Manufacturing and Operations industry, where organizations manage complex procurement, production, inventory, sales, finance, and workforce processes. Replacing separate systems and manual documentation with one integrated platform removes communication gaps and duplicate data entry, improving productivity, consistency, and transparency.

### 2.2 Existing Business Challenges

- Data Silos — disconnected Sales, Inventory, Purchasing, and Finance records causing inconsistent information.
- Manual Processes — attendance, payroll, purchase orders, invoicing, and stock updates done manually, increasing errors.
- Lack of Real Time Visibility — outdated manual reports, no live dashboards for management.
- Inefficient Communication — reliance on email/printed documents delays order processing and procurement.

### 2.3 Opportunities for Improvement

- Automated cross department workflows and elimination of duplicate data entry.
- Centralized database improving communication and reducing manual errors.
- Faster order processing, real-time reporting, and KPI dashboards.
- Data-driven management decisions and improved customer satisfaction.

## 3. ERP Solution Design — Module Overview

NexaERP consists of eleven integrated modules functioning as a complete enterprise management solution:

| Module             | Key Features                                                             | Stakeholders                                |
|--------------------|--------------------------------------------------------------------------|---------------------------------------------|
| Dashboard          | Real-time KPIs, sales/financial summaries, inventory & production status | CEO, CFO, Managers                          |
| Finance            | Chart of Accounts, Journal Entries, AP/AR, Budgeting                     | Accountants, Finance Managers, Auditors     |
| Sales              | Customer Management , Quotations, Sales Orders, Invoicing                | Sales Reps, Billing Officers                |
| Purchasing         | Vendor Management , Requisitions, Purchase Orders, Invoices              | Procurement & Purchasing Managers           |
| Inventory          | Product Management, Warehouses, Stock Levels & Movements                 | Warehouse & Inventory Controllers           |
| Manufacturing      | BOM, Work Orders, Production Orders, Quality Control                     | Production Engineers, Plant Managers        |
| Human Resource     | Employee Records, Attendance, Leave, Payroll, Recruitment                | HR & Payroll Officers                       |
| CRM                | Leads, Opportunities, Activities, Marketing Campaigns                    | CRM Plant Managers, Marketing Execs         |
| Project Management | Projects, Tasks, Time Tracking, Milestones                               | Project Plant Managers, Team Leaders        |
| Asset Management   | Asset Registration, Depreciation, Maintenance, Transfers                 | Asset Plant Managers, Maintenance Engineers |
| System Admin       | User Management, Roles & Permissions, Audit Logs, Settings               | System Admins, IT Support                   |

#### 4. Cross-Module Integration Workflow

Information entered in one module automatically updates related modules, removing duplicate entry:



HR data underpins all modules, dashboards give live analytics, inventory changes affect production planning, sales and purchases update financial records and supplier balances instantly.

## 5. Technology Framework & Cloud Deployment

### 5.1 Technology Stack

- **Frontend:** React 18, TypeScript, Vite, and Tailwind CSS for a responsive, reusable UI.
- **Backend:** Java 17, Spring Boot, Spring Security (JWT), Spring Data JPA, Hibernate, Flyway, and Lombok for secure RESTful APIs.
- **Database:** PostgreSQL (Development) and Azure PostgreSQL Database (Production) with optimized relational design.
- **Deployment:** Vercel for frontend hosting and Microsoft Azure App Service for backend deployment, with Azure PostgreSQL Database and Azure Blob Storage for cloud data and file management.
- **Version Control:** GitHub with branch management, pull requests, code reviews, and CI/CD support.

### 5.2 Azure Cloud Deployment

- **Vercel** hosts the React frontend, providing fast global content delivery, automatic deployments, and optimized performance.
- **Azure App Service** hosts the Spring Boot backend, delivering secure, scalable, and reliable API services.
- **Azure PostgreSQL Database** stores production data with high availability, automatic backups, and built-in security.
- **Azure Blob Storage** manages invoices, reports, product images, and other business documents.

**Benefits:** High availability, scalability, secure authentication, automatic deployment, cloud backup, disaster recovery, and remote accessibility

### 5.3 GitHub Version Control

Development follows a branch workflow with regular commits, pull requests, code reviews, database migration scripts, and project documentation, ensuring effective collaboration and high code quality throughout the development lifecycle.

## 6. Conclusion

NexaERP successfully streamlines manufacturing operations by integrating core business functions into a centralized ERP platform. Through automated workflows, real time dashboards, role-based access control, and cloud deployment, the system improves operational efficiency, enhances decision-making, and reduces manual effort and errors. Built using React, TypeScript, Vite, Tailwind CSS, Java 17, Spring Boot, Spring Security with JWT, PostgreSQL, Azure PostgreSQL Database, and Microsoft Azure, NexaERP provides a secure, scalable, and modern enterprise solution capable of supporting long term business growth and digital transformation.